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RESEARCH ARTICLE

Shifting tax incidence: Evidence from the Washington State cannabis market

Benjamin Hansen[1] | **Kendall Houghton**[2] | **Keaton Miller**[1] |
Caroline Weber[3]

[1]Department of Economics, University of Oregon, Eugene, Oregon, USA

[2]Center for Economic Studies, U.S. Census Bureau, Suitland, Maryland, USA

[3]University of Kentucky, Martin School of Public Policy & Administration, Lexington, Kentucky, USA

Correspondence
Caroline Weber, University of Kentucky, Martin School of Public Policy & Administration.
Email: caroline.weber@uky.edu

Abstract

We study how prices respond when a 25% gross receipts tax remitted by cannabis manufacturers was eliminated in Washington state and the retail excise tax was simultaneously increased from 25% to 37%. Standard theory suggests that this change should have increased welfare for manufacturers, retailers, and consumers; instead, our analysis shows that the reform unexpectedly shifted benefits toward manufacturers at the expense of retailers and consumers, who faced higher tax-inclusive prices post-reform. We hypothesize that this outcome was driven by behavioral factors such as anchoring and loss aversion. Our findings add to a growing body of evidence that tax reforms can affect market outcomes in ways not predicted by standard economic models, offering a cautionary lesson for policymakers considering similar reforms.

INTRODUCTION

Tax policy scholars often encourage policymakers to make decisions about tax reforms based on classical economic models. However, there is increasing evidence that prices respond to tax reforms in surprising ways that are not predicted by classical models, leading tax reforms to have unexpected first-order welfare consequences (Benzarti, 2024; Saez & Zucman, 2023). Identifying and understanding these nonstandard responses is critical for designing optimal tax reforms, as unexpected shifts in tax incidence can alter the intended distribution of costs and benefits among market participants.

We contribute to this discussion by analyzing a tax reform in Washington state's cannabis market. The reform was intended to simplify the tax structure and increase the competitiveness of the legal cannabis industry with the illicit market (Concerning comprehensive marijuana market reforms, 2015). Prior to the reform, a 25% gross receipts tax was levied at each stage of production: Producers remitted the tax when selling to manufacturers, manufacturers remitted the tax when selling to retailers, and retailers remitted the tax when selling to consumers. On July 1, 2015, the gross receipts tax was replaced with a single 37% retail excise tax. This reform increased production efficiency (Hansen et al., 2022) and our model of the industry predicts that it should have increased average welfare for all market participant types through a combination of price and quantity channels. However, our empirical analysis shows that the reform unexpectedly shifted benefits toward manufacturers at the expense of retailers and consumers, who faced higher tax-inclusive prices post-reform.

We study the reform using an interrupted time-series regression in first differences; that is, we ask how prices change in the week of the reform relative to the weeks surrounding the reform. Identification rests on the assumption that, after controlling for product characteristics and a baseline trend, prices would not have changed in the week of the reform if the reform had not occurred. We compare the results of this empirical analysis to a benchmark model of the industry that extends the model of Hansen et al. (2022) to include a retail sector. While the model predicts that manufacturers should have reduced their tax-inclusive prices by 25.3%, we find empirically that they only lowered prices by 7.2%.

We then examine retailer behavior. Our model predicts that retailers should respond to the reform by decreasing their tax-inclusive retail prices by 14.0%. We find that, instead, tax-inclusive prices *increased* by an average of 2.5%. In other words, retailers passed-through one third of the tax increase to consumers. Another roughly one third was borne by manufacturers, leaving retailers with about one third of the increase. We find evidence that retailers maintained constant tax-exclusive markups, which is consistent with our model's pricing rule conditional on manufacturers' behavior.

Our setting does not admit the explanations for shifting tax incidence previously found in the literature. Tax evasion is difficult throughout the supply chain due to the frequently-audited comprehensive regulatory reporting system (see, e.g., Slemrod, 2008). Tax rates in this setting are likely salient (see, e.g., Chetty et al., 2009), as the reform we study involves changes to taxes paid between firms, and agents on both sides of the market are simultaneously experiencing changes to their tax rate. Furthermore, prices change often and some of these changes are as large or larger than the predicted price response, so price rigidities cannot explain the result (see, e.g., Saez et al., 2012).

We hypothesize that our results are driven by behavioral factors, including anchoring and loss aversion. In short, manufacturers could gain substantial additional profits by anchoring themselves to their pre-reform prices and "doing nothing," whereas the same strategy would lead to losses if adopted by retailers. These possibilities, combined with the immediacy of the reform and its simultaneous application across the industry, may have helped manufacturers overcome coordination problems that they would otherwise typically face. We discuss these possibilities in more detail after presenting our results.

Our results have broader implications for tax policy beyond the cannabis industry. Many markets feature multistage or tiered tax structures, including other sin goods such as alcohol and tobacco, gasoline, and digital goods. Gross receipts taxes, once in decline, are making a comeback (Kaeding, 2017; Mikesell, 2007a, 2007b; Pogue, 2007; Testa & Mattoon, 2007). Our findings suggest that firms and consumers respond to the elimination of a gross receipts tax in non-standard ways, and thus reforms moving toward or away from a tiered tax structure may have unintended distributional consequences that policymakers should consider.

BACKGROUND

We analyze the adult-use cannabis market in Washington state, which opened in July 2014. We have written elsewhere about the history of this market (Hansen, Miller, Seo, & Weber, 2020; Miller & Seo, 2021); here we focus on features of the market and the reform underpinning our analysis.

The market consists of three types of firms: cultivators, who grow cannabis plants; manufacturers, who transform plant material into marijuana products; and retailers, who sell products they obtain from manufacturers to consumers. Potential entrants have to pass background checks and undergo a lengthy regulatory process requiring substantial capital investment before entry. Cultivators face capacity constraints—the largest firms may cultivate 30,000 square feet of plant canopy and may not merge to increase capacity. While retailers must be financially independent from other firms, a cultivator and a manufacturer may vertically integrate. When the reform was implemented, approximately 94% (by weight) of usable marijuana—dried and cured cannabis flowers—was produced through a vertically-integrated process (Hansen et al., 2022). Thus, we focus our analyses on two types of firms, "manufacturers" and "retailers."[1]

The market is closed: All cannabis sold by retailers is grown in the state and every ounce grown legally within the state is sold at a Washington retailer. These rules are enforced through the state's "seed-to-sale" traceability system, which tracks each plant from cultivation through processing and retail. The collected data can be used to check for tax evasion: Retailers cannot sell cannabis without manufacturing records, which forces manufacturers to report accurately.[2] Reporting is enforced through frequent audits—firms typically face one or more visits per year—backed by significant penalties.[3]

Washington's initial tax regime consisted of a 25% tax collected at every transfer of cannabis. Vertically-integrated manufacturers owed no tax on intra-firm transfers. The reform we analyze eliminated the 25% excise taxes within the supply chain and increased the retail excise tax rate to 37%. The excise tax at retail applied to the sales-tax-inclusive price pre-reform and the sales-tax-exclusive price post-reform. Accounting for changes to the base and rate of the retail tax, the reform changed the average retail tax rate by 6.93%.[4] This change was designed to be revenue neutral under the assumption that tax-*exclusive* prices remained constant (whereas tax invariance predicts constant tax-*inclusive* prices). We account for both the change in the rate and the base of the retail excise tax in our analyses. We provide calculations of revenue pre- and post-reform in the section "Data and Methods." Other regulations concerning cannabis production, distribution, and sales were unaffected.

Our identification assumes that the policy change was unanticipated. HB 2136—the bill that made the changes we study—originated and was passed in the Washington House during the 2015 Regular Session but stalled in the Senate. It was crafted to change the tax structure to a single excise tax at retail, to increase the tax rate at retail, and to make some changes regarding how the revenue would be spent. The proposed changes in tax structure were generally the least controversial part of the bill; the motivation for these, as stated in the bill, was to increase the competitiveness of the legal market with the illicit market (Concerning comprehensive marijuana market reforms, 2015). The Republican-led house and Democrat-led Senate had different views on the appropriate tax rate to impose on cannabis retailers and what to spend the money on. Democrats wanted to largely preserve the original spending allocations in I-502, while Republicans preferred to use the funds to cover education spending required by the McCleary Supreme Court decision (Bush, 2015). These disagreements left considerable uncertainty about its passage. The bill was reintroduced in the First Special Session but again stalled. Finally, on June 27 during the Second Special Session, the bill passed both chambers. This was one of several pieces of legislation passed as legislators crafted a budget for the 2015 to 2017 biennium to keep the state government from shutting down on July 1. The Governor signed HB 2136 on June 30 and the law went into effect the next day. Contemporaneous reporting portrayed the industry as unprepared. According to one retail manager, "[we] have a few hours to change an entire market's pricing structure. It is an exceptionally short window for such a tremendous change" (La Corte, 2015).

[1] State law calls cultivators "producers" and manufacturers "processors"—we choose nomenclature to represent functional equivalents in other markets.

[2] Retailers can underreport sales, but such behavior is detectable by comparing retail sales and wholesale purchases. Our estimates are unaffected by dropping the few retailers that underreport substantially.

[3] We observe audit violations and our estimates remain approximately the same if we drop firms with underreporting violations. Moreover, our results are not consistent with asymmetric tax evasion. Manufacturers would pass along less tax savings if retailers are more effective tax evaders, but in that case, retailers would still not pass along any of the tax to consumers.

[4] The average sales tax rate during this period was 8.9%, thus $\log(\frac{1.25(1+0.089)}{1.37+0.089}) = -0.0693$.

BENCHMARKING MARKET RESPONSES

In this section, we develop benchmark responses to the tax reform against which we compare our empirical findings. We do so by extending the model, calibration, and prediction exercise of Hansen et al. (2022), which we refer to as HMW, to include retail firms. In the interest of brevity and as the model is largely identical to that work, we focus here on our changes and calibration results. Mathematical details are left to Appendix C.[5]

A brief summary of HMW

HMW present a tractable supply chain model in the style of Weyl and Fabinger (2013). In the model, firms may engage in either or both of two activities: producing intermediate goods from raw materials (which in this context refers to the growth and harvesting of plant material) and processing intermediate goods into final goods (which in our context refers to the processing plant material into usable cannabis products). There are two markets: a market for intermediate goods and a market for final goods. As is common in double-marginalization problems, the demand for final goods is taken to be exogenous while demand for intermediate goods is endogenous. Within each market, goods are assumed to be identical, though firm behavior in each market is characterized by an exogenous firm-specific *conduct parameter*: Under the *conduct condition*, firms set prices to equate their elasticity-and-tax-adjusted Lerner index equal to their conduct parameter.[6]

HMW calibrated the model using moments derived from pre-reform data and used it to benchmark post-reform outcomes. They concluded that the model is able to capture outcomes in the market "remarkably close[ly]" (Hansen et al., 2022, p. 19). For example, the model predicts that the firms engaged in selling intermediate goods would increase their production of those goods by 30.6%; the empirical estimate is 31.2%.

Extension and calibration

We extend HMW by adding a retail sector which is subject to excise taxes. This is straightforward in this setting because Washington's regulations require that retail firms be owned and operated entirely independently from other firms; in other words, we can add retail firms to the model without affecting the equilibrium conditions in the intermediate goods market. As detailed in Appendix C, imperfect competition in the retail sector is captured by a retail conduct parameter. This parameter links retail prices to intermediate goods prices via an additional equilibrium condition. The resulting model features a tax invariance (TIV) result: If, instead of an excise tax paid by retailers, consumers paid a sales tax, the equilibrium conditions, and therefore outcomes, would be identical.[7] We proceed by calibrating the model following HMW: The calibration environment features two upstream firms who face quadratic costs and one retailer. We use the

[5] All appendices are available at the end of this article as it appears in JPAM online. Go to the publisher's website and use the search engine to locate the article at http://onlinelibrary.wiley.com.

[6] Weyl and Fabinger (2013) showed that when products are weak substitutes, the conduct parameter in models of profit maximization ranges from 0 (perfect competition) to 1 (monopoly).

[7] As is common in this literature (see e.g., Mace et al., 2020), we assume a constant-price-elasticity functional form for demand. This results in a demand system that is log-convex. As discussed by Bagnoli and Bergstrom (2006) and Weyl and Fabinger (2013), this curvature assumption restricts the possible tax pass-through rates of the model; pass-through rates may exceed unity if and only if demand is log-concave. However, our calibration results, our empirical results, and the results of others who have studied this market and similar markets suggest that pass-through rates are less than 1 (Conlon & Rao, 2015; Hollenbeck & Uetake, 2021; Huang et al., 2021; Mace et al., 2020; Miller & Seo, 2021) so this restriction is likely nonbinding. Changing the assumption of constant-elasticity to constant-curvature would not generate the kind of responses that we ultimately estimate because such a model would still feature a "tax invariance" result and our results imply a violation of tax invariance.

TABLE 1 Calibration summary.

(a) Parameters

Variable	Description	Value
ε_D	Demand elasticity	1.1689
k	Demand level	449.477
β_{g1}	Cost to firm 1 of cultivating cannabis	0.0771
β_{g2}	Cost to firm 2 of cultivating cannabis	0.0429
β_{p1}	Cost to firm 1 of processing cannabis	0.0693
β_{p2}	Cost to firm 2 of processing cannabis	0.0254
θ_p	Conduct parameter in wholesale market	0.1447
θ_r	Conduct parameter in retail market	0.6087

ε_D and k are calibrated from Hansen et al. (2020). The other parameters are calibrated using the below moments.

(b) Pre-Reform Moments

Moment	Data	Model
Fraction vertically integrated	0.955	0.947
Manufacturer to retailer price	3.60	3.60
Cultivator to manufacturer price	1.53	1.53
Firm 1 / Firm 2 cultivation ratio	0.413	0.418
Firm 1 / Firm 2 processing ratio	0.537	0.534
Retail price	9.39	9.39

See text for detailed description of moments.

(c) Post-reform predicted outcomes

Outcome	Prediction
Change in log-manufacturer price	
Tax-exclusive	−0.198
Tax-inclusive	0.090
Change in log-retail price	
Tax-exclusive	−0.198
Tax-inclusive	−0.144

Note: We model the reform, which consisted of a change in both the base and the rate of the retail tax, by removing the gross receipts tax on manufacturers, and increasing the retail tax by 6.93%.

moments of HMW with the addition of the average tax-exclusive price-per-gram charged by retailers to consumers in the pre-reform period. The results of our calibration are reported in Table 1, which is analogous to Table 3 of HMW. Panel (a) reports the calibrated values of the parameters, while Panel (b) reports the moments in the data and in the calibrated model. Just as in HMW, the model can closely match the moments: Each model moment is within approximately 1% of the data.

Panel (c) reports the prices changes predicted by the model in a form that we can take to the data. The predicted change in the tax-exclusive log-manufacturer price (that is, $\ln(p_f^{post}) - \ln(p_f^{pre})$) is −0.198. That is, the model predicts that the reform should reduce the prices paid by retailers to manufacturers by approximately 20%. Net of taxes, the price charged by manufacturers increases slightly. The predicted change in the log-retail price (that is, $\ln(p_r^{post}) - \ln(p_r^{pre})$) is −0.144, implying

TABLE 2 Pre-reform retail summary statistics.

Variable	Obs.	Mean	Std. Dev.
Prices and Taxes			
Tax-Inclusive Retail Price ($/g)	63,668	13.033	3.798
Tax-Exclusive Retail Price ($/g)	63,668	9.570	2.783
Manufacturer Price ($/g)	63,668	4.103	1.309
Retail State + Local Sales Tax Rate	63,668	1.089	0.006
Tax Revenue Pre-Reform ($/g)	63,668	4.489	1.246
Competition			
Market Share of Retailer in 10 Mile Radius	63,668	0.313	0.282
Market-level Manufacturer Market Share	63,668	0.014	0.016
Retail-Level Manufacturer Concentration Index	63,668	6.997	2.691
Benchmarks Assuming TIV			
Expected Tax Revenue Post-Reform ($/g)	63,668	4.104	1.200
Manufacturer Pass-Through Cents	63,668	−0.640	0.185
Manufacturer Pass-Through Percent Change	63,668	−0.177	0.058

Notes: An observation is an inventory-lot-week pre-reform. The data come from our retail analysis set and cover the 6 weeks prior to the tax reform. Tax revenue is calculated using both excise and state and local sales taxes. The retail-level manufacturer concentration index is calculated as follows: for a given retailer, we sort their suppliers by the weight of inventory sold, and count the number needed to comprise at least 75% of total sales. The "benchmarks assuming TIV" account for changes in the base and rate of the retail excise tax. The "manufacturer pass-through" statistics assume constant tax-inclusive retail prices and indicate the post-reform changes to manufacturer prices that would have left retailer variable-profit-per-gram constant. These probabilities are calculated for the subset of retail-processor-strain-weight group-weeks when the inventory lot changes (and thus a new purchase from a manufacturer has occurred).

that the reform should reduce retail prices paid by consumers by approximately 15%. Exclusive of taxes, retailers should decrease their prices by roughly 20%.

When combined with HMW's result that the reform increased quantities, these results imply that, on average, all categories of market participants should benefit from the reform. Manufacturers should gain through the quantity channel and the price channel, retailers should gain through the quantity channel (as the model predicts essentially constant variable profit per unit sold), and consumers should gain through both channels.

DATA AND METHODS

Our data consist of records from the traceability system maintained by the Washington State Liquor and Cannabis Board. We obtain data on all plants, products, and sales. We restrict our analysis to "usable marijuana" products[8]—74.5% of the total transactions observed in our data. Within this category, products are differentiated by strain (analogous to fruit cultivars), potency, and form-factor (loose or pre-rolled). These characteristics are captured by our fixed effects.

We aggregate retail sales by inventory-lot-week, where an inventory-lot is a batch of identical product. We exclude firms with less than 2 months of pre- and post-reform data. The reform also changed technical reporting rules which affect the price data. We clean the data to reflect the prices faced by

[8] *Useable marijuana products* refers to cannabis flowers which have been dried, cured, and packaged into sealed packets containing a set weight (e.g. 1 gram, 3.5 grams, or 7 grams).

TABLE 3 Manufacturer price response.

	(1) Δlog(Price)	(2) Δlog(Price)	(3) ΔPrice	(4) Δlog(Price)
		Tax Reform		
ΔTax Reform	−0.065***	−0.072***	−0.228***	−0.059***
	(0.015)	(0.018)	(0.068)	(0.014)
Observations	12,087	12,087	12,087	34,085
Manufacturer Firms	199	199	199	380
P-Value for Test of Predicted Pass-Through	0.000	0.000	0.000	0.000
		Placebo		
ΔPlacebo	0.001	0.000	0.014	−0.002
	(0.012)	(0.014)	(0.040)	(0.010)
Observations	16,302	16,302	16,302	80,677
Manufacturer Firms	175	175	175	579
Bandwidth	6 weeks	6 weeks	6 weeks	6 months
MRS FE?	No	Yes	Yes	Yes
Aggregation	Weekly	Weekly	Weekly	Monthly

Notes: This table reports estimates of Equation (1)—other variables in that equation are included, but not reported. An observation is a manufacturer-retailer-strain-week. The outcome is the change in the log of the price per gram charged by the manufacturer to the retailer (except for in column (3) which is the same outcome, but not logged). MRS stands for manufacturer-retailer-strain fixed effects. The estimates are weighted by the total grams sold across the two weeks of the difference. The P-value tests the null hypothesis that the estimated pass-through is equal to that predicted by TIV. For the placebo regressions, we repeat the analysis one year later. These regressions are estimated with reghdfe in Stata. Standard errors two-way-clustered by manufacturer and retailer are in parentheses (Cameron et al., 2011). *5% significance level. **1% significance level. ***0.1% significance level.

consumers using an algorithm based on retail rounding behavior verified by spot checks of historical menus.[9] See Appendix B for details.

Table 2 reports summary statistics for retail lots for the 6 weeks pre-reform (the basis for our analyses in the section "Results"). The average tax-inclusive retail price was $13.03 per gram and the tax-exclusive price was $9.57 per gram. Retailer tax-exclusive prices are more than double manufacturer tax-inclusive prices.

The average market share of retailers in the 10-mile radius around their location was 31%, suggesting that there is substantial retail market power (Hollenbeck & Uetake, 2021; Mace et al., 2020). The manufacturer market is effectively state-wide; the average market share is 1.4% and none exceeds 7%. However, manufacturers may exert market power on individual retailers. We construct a retail-level manufacturer concentration index by sorting each retailer's suppliers by the weight of inventory sold and counting the number of manufacturers that comprise at least 75% of sales. On average, seven manufacturers supply 75% of a retailer's inventory.

We plot the panel of retail tax-exclusive prices normalized to the week before reform in Figure 1. For each week, we take inventory lots in their first week of sale and match them with the price paid to the manufacturer, restricting observations to those where the first retail sale and manufacturer sale both happened pre- or post-reform; thus, this illustrates the relation between retailer per-gram revenue and variable costs. The two series move in a highly correlated way through the entire pre- and post-reform period (including the period around April 20, an industry promotional event). This implies a constant markup of the retail tax-exclusive price over variable costs (the manufacturer price) that appears to be preserved in response to the tax reform. This figure does not suggest that there is a longer-run response

[9] Cannabis retailers set tax-inclusive prices that are round numbers (e.g., $8 or $10.25). While this represents a potential friction, the effective minimum price change is smaller than the effects we estimate.

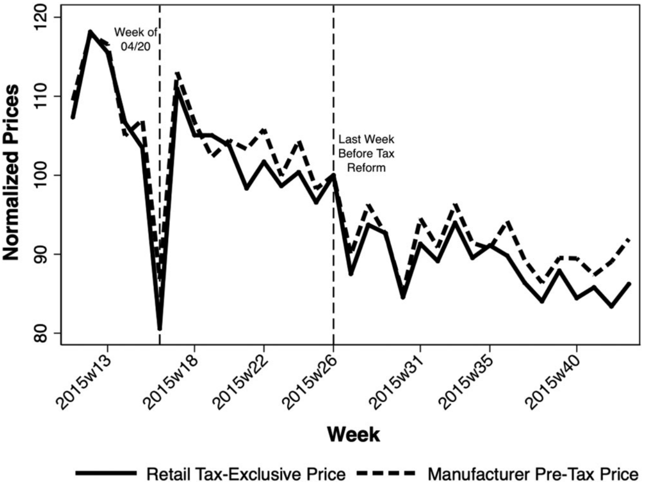

FIGURE 1 Matched retail and manufacturer prices.

Notes: This figure plots raw average prices in Washington's cannabis industry for 4 months before and after the tax reform, normalized to 100 in the week before the reform. For each week, we take inventory lots in their first week of sale and match them with the price paid to the manufacturer, restricting observations to those for which the first retail sale and manufacturer sale both happened pre- or post-reform (before any applicable taxes are paid from the manufacturer to the government). This therefore illustrates the relation between retailer per-gram revenue and variable costs. The left dashed line in the figure marks 4/20 (an industry promotional period) and the right dashed line marks the week before the tax reform.

to the reform. If there were a slow-moving adjustment towards the price adjustment predicted by our model, the gap between matched wholesale and retail prices (which will both experience the same market level shocks) would slowly diverge as wholesale prices continued to adjust downwards relative to retail prices towards the equilibrium price adjustments predicted in Table 1. If anything, we see the opposite—wholesale prices are rising slightly relative to retail prices in the period post-reform. This figure depicts a set of products that is changing over time. To disentangle the effects of the reform from long-run trends and control for potential compositional changes, we employ regression and, in the section "Event Studies," event study analyses.

Empirical specification

We model responses to the tax reform as an interrupted time-series in first differences:

$$\Delta \log (p_{it}) = \alpha_0 + \alpha_1 \Delta TaxReform_t + \alpha_2 FE_i + u_{it} \tag{1}$$

where i is the unit of observation as described for the manufacturer and retail analyses below, and t indicates the week. p is the wholesale or retail price per gram, *TaxReform* is an indicator variable that is one after July 1, 2015, and zero before, and FE are fixed effects. α_1 is the parameter of interest.[10] Because all firms are treated and generally included in our analysis, this parameter is an ATT. Our analysis window spans 6 weeks before and after the reform.[11] We two-way cluster standard errors on manufacturer and retail location (Cameron et al., 2011).[12] Our identifying assumption is that within a given product, there are no shocks in the week of the reform that would have a significant and

[10] Without fixed effects, this regression is equivalent to an interrupted time series regression in levels with fixed effects at the level of our first differences and a control for time to the reform.

[11] We demonstrate our estimates are robust to this choice in Appendix Figure A.5.

[12] Firm clusters or two-way clusters on firm and week yield similar standard errors.

systematic impact on prices besides the direct effect of the tax reform. Given the short interval between observations (i.e., a week, not a year), this assumption is plausible.

For the manufacturer analysis, we aggregate to the manufacturer-retailer-strain-week level, so that i is a manufacturer-retailer-strain tuple, and then take first differences.[13] Each manufacturer-retailer-strain tuple does not sell every week. We thus calculate the minimal-length difference and include difference-length fixed effects.[14] The maximum difference-length allowed is 4 weeks. We are thus estimating the magnitude of price changes in response to the reform within a specific firm-product pairing. Because our dependent variable is in first-differences, retailer-manufacturer-strain fixed effects allow each retailer-manufacturer-strain price to have a separate time trend.

For our main retail analysis, we aggregate to the inventory-lot-week level so that i is a retail inventory lot. Retail sales from a given inventory lot are frequent, so we construct 1-week differences. We are thus estimating the change in the retail price of an inventory lot in response to the tax reform holding all possible product and firm variation constant. Sales of retail inventory lots typically last multiple weeks, so we include fixed effects for weeks from the first sale out of that inventory lot. Because our dependent variable is in first-differences, inventory-lot fixed effects allow prices in each inventory-lot to have a separate time trend.

We separately examine the first week of retail sales for each inventory-lot and include only those that were purchased from manufacturers in the same week. Similar to our manufacturer analysis, we aggregate by retailer-manufacturer-strain and take varied length differences. We include difference-length fixed effects. In these regressions, we ask how prices for *new inventory-lots purchased post-reform* change relative to *pre-reform lots of the same strain from the same manufacturer*. This allows us to examine whether prices change more or less if the inventory was purchased post-reform relative to inventory that had already been purchased and was selling pre-reform.

Our identifying assumption is much more likely to hold in our setting than the assumption made in a classic difference-in-differences design—that, for a lengthy period (e.g., a year), two states would experience the same systematic price shocks in the cannabis market. In general, it's not clear which of these two would be more likely to hold a priori, but in this setting, it is very unlikely that the latter would hold because each state is a closed market and there are substantial differences in market regulatory structures, processes, and outcomes. Moreover, we will provide evidence through placebo analyses and event studies that we have no reason to reject the null hypothesis that our identifying assumption is valid. When we use the same data to examine another natural experiment that allows for the addition of a comparison group to our regression discontinuity in time (RDIT) design[15] (i.e., a difference-in-discontinuities regression; Hansen, Miller, & Weber, 2020), we found the estimates remained quantitatively and qualitatively similar.

Our implementation of interrupted time series, also known as RDIT, addresses critiques previously raised against this method (Hausman & Rapson, 2018). We select a narrow bandwidth (measured in weeks, not years) and we estimate the regression in first-differences rather than levels; this, along with our event study figures, allows us to precisely pin down the response in the week of the reform, rather than allowing the regression to obtain part of its identification from shifts that may happen many weeks away from the reform. In addition, we address autocorrelation over time with firm-level standard error clustering, we aggregate at the weekly level to avoid challenges associated with estimating day-of-week fixed effects, and we include fine-grained fixed effects to address any compositional shifts over time.

[13] Aggregation beyond the inventory lot is required because each lot is sold only once. Other possible aggregations produce similar estimates with lower power (though statistical significance remains).

[14] These fixed effects are not significant. Our estimates are similar when restricted to 1-week differences, but with less power.

[15] Washington firms near the Oregon border were "treated" while those away from the border were not.

RESULTS

Manufacturer results

Table 3 reports estimates of equation (1) for manufacturers. Column 1 estimates a 6.5% price decline in response to the tax reform. When we add manufacturer-retailer-strain fixed effects in column 2—our baseline specification—the point estimate becomes −7.2% (p-value = 0.000). This is roughly one quarter of the 19.8% decrease predicted by our calibrated model in the section "Extension and Calibration" and we can reject the null hypothesis that these estimates are the same (p-value = 0.000). Column 3 repeats column 2 for the price in levels instead of logs—the reform decreased manufacturer prices by 23 cents, again about one third of what is predicted by our calibrated model. Table 3 column 4 aggregates the data by months instead of weeks and we find the estimates are very similar with smaller standard errors. This aggregation is an alternative way to address manufacturers not selling every strain they produce to every retailer every week and also allows us to examine a longer-term response.

The bottom panel of Table 3 repeats the specification of each column for a placebo reform dated 1 year later. The estimates are near zero across all four columns, providing support that our regression specifications are valid.[16] One could subtract the placebo estimates from the main estimates to create a differences-in-RD design; the estimates would be very similar. It is possible that our rejection of the price change predicted by our calibrated model is driven by price stickiness or that the dampened response we observe (relative to what our framework predicts) is driven by the sheer magnitude of our predicted response relative to typical price shifts in this market. Figure 2 provides evidence of price mobility by plotting the entire distribution of weekly price changes for each retail-manufacturer-strain pair in the baseline estimate as a histogram in the top panel. The period of the tax reform is marked by the hollow green histogram and surrounding periods are marked by the gray histogram. The width of each bin is 0.04, so all price changes within 2% of zero are included in the bin centered around zero. The dashed lines from right to left indicate full pass-through predicted by models of firm behavior (−25.3%) and zero (i.e., no price change). We repeat this histogram for monthly price changes in Appendix Figure A.1. The bottom panel of Figure 2 is a time series plot of the weekly likelihood of retail-manufacturer-strain pair price changes of more than 2%.

Figure 2 and Appendix Figure A.1 illustrate several things. First, we see the distributional shift in the period of the reform in the histograms. Note that there is a substantial increase in pass-through all the way down to what our calibrated model predicts. A visually detectable increase extends slightly further down to full pass-through of the tax savings onto retail firms. While the aggregate response is a sizable shift in incidence from manufacturers to retailers, there are retail-manufacturer-strain tuples for which our model predictions appear to hold in isolation. About 14% of manufacturer-retail-strain pairs are within 10% of what our calibrated model predicts. Splitting the manufacturer tax break evenly in half is not predicted by the model but is more common in practice—almost 20% of manufacturer-retail-strain pairs are within 10% of this choice. Second, the histograms highlight that large price shifts do occur with reasonable frequency in the absence of the reform—about 14% of retail-manufacturer-strain tuple weekly price decreases were at least as large as predicted by our model (this doubles in the period of reform). This statistic rises to 17% at the monthly level. Moreover, price changes are frequent in general; both panels of Figure 2 show that about 70% of retail-manufacturer-strain tuples adjust their prices by at least 2% before the reform and this increases to 85% in the period of the reform. Even if we rescaled our estimate assuming that any observation with minimal adjustment in

[16] In addition, we estimate many placebo regressions and construct a placebo permutation test for manufacturers in the top panel of Appendix Figure A.4. There are no placebo estimates as extreme as our estimate and the implied p-value is 0.012.

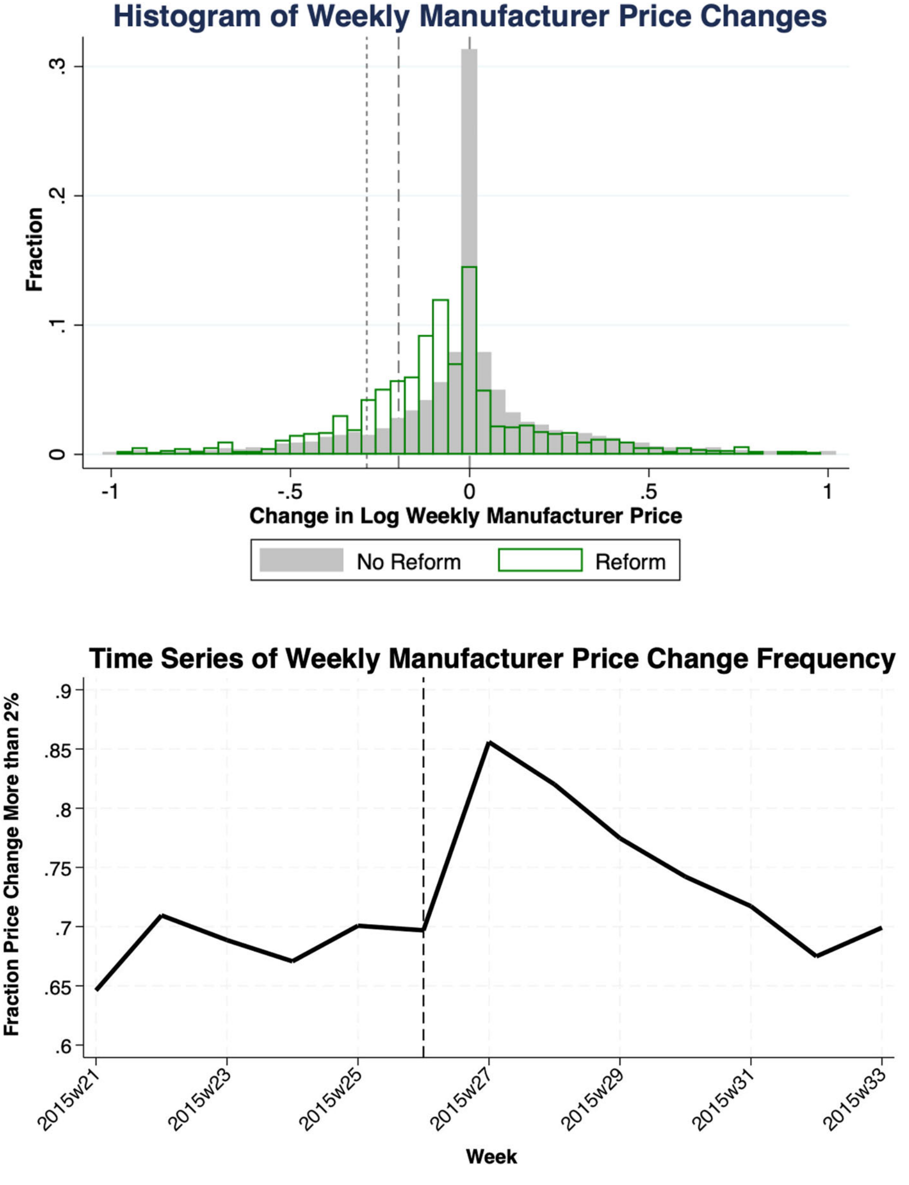

FIGURE 2 Histogram and time series of manufacturer price changes.

Notes: The top panel plots the dependent variable, $\Delta log(Price)$, for each retail-manufacturer-strain tuple in the baseline weekly estimate sample. The period of the tax reform is marked by the hollow green histogram and the other surrounding periods—6 weeks pre- and post-reform—are marked by the gray histogram. The width of each bin is 0.04, so that all price changes within 2% of zero are included in the bin centered around zero. The dashed lines from left to right indicate: full pass-through (−28.7%), pass-through predicted by our calibrated model (−19.8%), and zero (i.e no price change). Outliers outside the interval [−1,1] are excluded. The bottom panel plots the likelihood of weekly price changes for each retail-manufacturer-strain tuple of 2% or more for the baseline estimate sample for the 6, weeks before and after the reform. The dashed line marks the week before the tax reform.

this period was caused by rigidities or lack of awareness, the data would still reject our calibrated model predictions.[17]

[17] However, this does not rule out additional adjustments by many firms over a longer horizon. Indeed, the bottom panel of Figure 2 suggests that there could be some longer-term adjustments as price changes are more common for several weeks after the reform. To fully assess whether there is evidence of a longer-run price adjustment towards the calibrated model prediction, we consider event studies in the section “Event Studies.”

TABLE 4 Manufacturer price response heterogeneity.

	(1)	(2)	(3)	(4)
	Tax Reform			
ΔTax Reform	−0.071***	−0.058**	−0.072***	−0.058**
	(0.018)	(0.020)	(0.019)	(0.020)
ΔTax Reform x Whole Price Rounding	−0.020			
	(0.038)			
ΔTax Reform x Price Deviation from Mean		−0.083***		−0.084***
		(0.020)		(0.021)
ΔTax Reform x Manufacturer-Retailer Share			−0.024	−0.011
			(0.017)	(0.016)
Observations	12,087	8,445	11,212	7,882
Manufacturer Firms	199	183	193	182
	Placebo			
ΔPlacebo	0.002	0.001	0.000	−0.008
	(0.016)	(0.016)	(0.020)	(0.019)
ΔPlacebo x Whole Price Rounding	0.018			0.016
	(0.031)			(0.036)
ΔPlacebo x Price Deviation from Mean		−0.012		−0.010
		(0.012)		(0.013)
ΔPlacebo x Manufacturer-Retailer Share			0.030	0.034
			(0.058)	(0.058)
Observations	16,302	11,235	15,345	10,677
Manufacturer Firms	175	154	169	153
Bandwidth	6 weeks	6 weeks	6 weeks	6 weeks
MRS FE?	Yes	Yes	Yes	Yes
Aggregation	Weekly	Weekly	Weekly	Weekly

Notes: This table reports estimates of Equation (1) with additional covariates interacted with ΔTax Reform—other variables in that equation are included, but not reported. An observation is a manufacturer-retailer-strain-week. The outcome is the change in the log of the price per gram charged by the manufacturer to the retailer (except for in column 3 which is the same outcome, but not logged). Whole Price Rounding is a measure for whether the manufacturer price per gram is rounded to the nearest whole dollar. Price Deviation from the Mean measures the mean gap between the manufacturer-retailer-strain price pre-reform and the average pre-reform leave-one-out strain price. Manufacturer-Retailer Share is the pre-reform share of the manufacturer's sales going to a specific retailer. MRS stands for manufacturer-retailer-strain fixed effects. The estimates are weighted by the total grams sold across the two weeks of the difference. For the placebo regressions, we repeat the analysis one year later. These regressions are estimated with reghdfe in Stata. Standard errors two-way-clustered by manufacturer and retailer are in parentheses (Cameron et al., 2011). ∗5% significance level. ∗∗1% significance level. ∗∗∗0.1% significance level.

Figure 2 is consistent with our identification assumptions and therefore provides additional evidence of the validity of our regression estimates—the price shifts are concentrated in the region one would expect and, while price shifts are common in this market, their frequency and magnitude increases in the period of reform. That is, there are substantial increases in the distribution of price decreases in the period of the reform in each of the bins up to what is predicted by our framework and much less beyond that. As expected, this can be seen most cleanly in the weekly price change histogram. If the responses we estimate were partially attributable to some more generic market shift in that same week, there is no reason to think that the price shifts would have this particular pattern.

We consider three possible explanations for the heterogeneity we observe in Figure 2 in Table 4: rounding, price corrections, and bargaining power. We do this by interacting covariates that measure

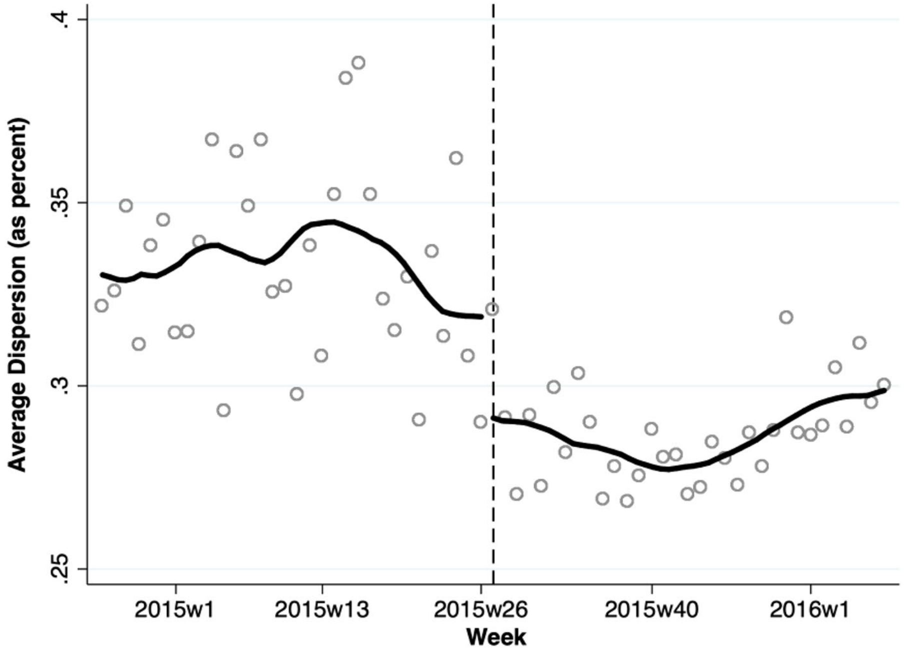

FIGURE 3 Strain price dispersion.

Notes: This figure illustrates shifts in price dispersion in this market across time. We define price dispersion here as the weekly average absolute difference between a leave-one-out average strain price and the price of each transaction. Dispersion declines by 3.3 percentage points after the tax reform (p-value = 0.045).

these possibilities with $\Delta TaxReform$. In column 1, we explore the impact of whole-price rounding: Some manufacturers round their prices, which may affect their ability to adjust prices in response to the reform.[18] Our measure of rounding is simply an indicator variable for whether the manufacturer rounded its prices to the nearest dollar prior to the reform. Next, tax rate changes could potentially be used as an event to coordinate price adjustments to decrease the dispersion of prices in the overall market; we examine whether those with pre-reform manufacturer-retailer-strain prices that are further away from the mean leave-one-out strain price make larger adjustments in column 2. Lastly, bargaining power could affect how firms respond and this may be correlated with the observed price dispersion prior to the reform, so we examine whether the pre-reform share of the manufacturer's sales going to a specific retailer matters in column 3. In column 4, we combine these three variables into a single regression.[19] We find that rounding and bargaining power do not matter. The pre-reform gap between the average manufacturer-strain price charged to a retailer and the average leave-one-out strain price does matter: A one standard deviation increase in the pre-reform manufacturer-retailer-strain price relative to the average strain price more than doubles the downward price adjustment in response to the reform (and the opposite is true for below average pre-reform manufacturer-retail-strain prices). This does not explain the unexpected shift in incidence away from manufacturers, on average, but does illustrate an important mechanism for heterogeneous firm decisions. Moreover, as depicted in Figure 3, this response decreases the overall price dispersion in the market by 3.3 percentage points (this is measured as the change in the absolute percent difference between the leave-one-out average strain price and the price of each manufacturer to retailer sale).

[18] Whole price rounding is likely adopted because of limited access to banking in this market (Berger & Seegert, 2024).

[19] Note that our measures of price deviation and manufacturer-retailer share have been standardized.

TABLE 5 Retail tax-exclusive price response.

	(1)	(2)	(3)	(4)	(5)	(6)
	Δlog(Price)	**Δlog(Price)**	**Δlog(Price)**	**ΔPrice**	**Δlog(Price)**	**Δlog(Price)**
			Tax Reform			
ΔTax Reform	−0.045***	−0.044***	−0.046***	−0.413***	−0.049**	0.011
	(0.006)	(0.007)	(0.006)	(0.065)	(0.018)	(0.017)
Δlog(Manufacturer Price)						0.887***
						(0.084)
Observations	145,357	145,357	145,357	145,357	11,265	11,265
Retail Firms	110	110	110	110	110	110
Implied Tax-Inclusive Price Change	0.024	0.025	0.023	0.230	0.020	0.080
P-Value for Test ofConstant Tax-Inclusive Price	0.000	0.000	0.000	0.000	0.270	0.000
			Placebo			
ΔPlacebo	−0.006*	−0.004	0.001	−0.029	−0.016	−0.004
	(0.003)	(0.003)	(0.002)	(0.017)	(0.012)	(0.009)
Δlog(Manufacturer Price)						0.642***
						(0.053)
Observations	253,123	253,123	253,123	253,123	11,534	11,534
Retail Firms	106	106	106	106	105	105
Bandwidth	6 weeks	6 weeks	6 weeks	6 weeks	6 weeks	6 weeks
MRS FE?	No	No	No	No	Yes	Yes
Inventory Lot FE?	No	Yes	Yes	Yes	No	No
Difference Length	1 week	1 week	2 weeks	1 week	1-4 weeks	1-4 weeks
Restricted to First Week Sales?	No	No	No	No	Yes	Yes

Notes: This table reports estimates of equation (1)—other variables in that equation are included but not reported. An observation is an inventory-lot-week. The outcome is the log of the tax-exclusive price per gram charged by the retailer to consumers (except for in column 4 which is the same outcome, but not logged). MRS stands for manufacturer-retailer-strain fixed effects. The estimates are weighted by the total grams sold in the first week of the difference. The P-value tests the null hypothesis that the tax-inclusive price remained constant as predicted by TIV. For the placebo regressions, we repeat the analysis one year later. These regressions are estimated with reghdfe in Stata. In the last two columns we only include observations in their first week of being sold at retail and only if the cannabis was also purchased from the processor in that same week. Standard errors twoway-clustered by manufacturer and retailer are in parentheses (Cameron et al., 2011). ∗5% significance level. ∗∗1% significance level. ∗∗∗0.1% significance level

Retail price results

Table 5 reports estimates of equation (1) for retailers. The coefficients are very similar in columns 1 and 2. The coefficient in column 2—our baseline specification which adds inventory lot fixed effects to column 1—implies that the reform reduced tax-exclusive retail prices by 4.4% (p-value = 0.000). Combined with the rate change, this implies that tax-*inclusive* prices increased by 2.5%; retailers passed through roughly one third of the tax to consumers. We can reject the price adjustment predicted by our calibrated model (p-value = 0.000).

As firms might have taken time to adjust (and the Independence Day holiday may have generated temporary price adjustments), column 3 repeats column 2 for 2-week differences and drops the first week after the reform, so that the effect of the reform is identified from the difference between the week before and the week after the reform. The estimates are approximately the same, indicating that neither of these concerns play a large role. We will return to a broader discussion of timing in "Event Studies."

Table 5 column 4 repeats column 2 with the dependent variable in levels—we estimate that average retail tax-exclusive prices fell by 41 cents per gram. Thus, retailers are an average of 41 cents per gram worse off on existing inventory as a result of the reform. On fresh inventory, retailers were roughly 18 cents per gram worse off (41 less the estimated 23 cent decrease in manufacturer prices estimated in Table 3). In other words, this reform caused smaller manufacturer price cuts than were predicted by the model, which shifted incidence away from manufacturers toward retailers. Moreover, our framework predicts consumers will be better off after the reform, but they are, in fact, worse off as they now face higher tax-inclusive prices.

Table 5 columns 5 and 6 take an alternative approach to identification examining inventory lots only in their first week and only if retailers purchased the inventory lot from the manufacturer in that week. For this, we create a panel of retail-processor-strain-weight group-weeks. The estimates are quite similar suggesting retailers' price responses are largely unaffected by whether they are selling inventory lots purchased pre-reform or selling new inventory lots purchased post-reform. Column 6 adds the first-differenced log manufacturer price; the coefficient on the wholesale price is not statistically different from one and the coefficient on $\Delta \mathit{TaxReform}$ is now approximately zero. This suggests retailers largely maintained a constant tax-exclusive markup.[20] This is consistent with the adjusted Lerner pricing rule of equation (2). Hence, while retail behavior as a whole is inconsistent with our calibrated model, after conditioning on the pass-through from manufacturers, retailer behavior is consistent with marginal-cost pricing.

In the bottom panel of Table 5 and in the bottom panel of Appendix Figure A.4, we repeat the placebo analyses we perform for our manufacturer sample for the retailer sample. These analyses provide no evidence of bias in our estimates.

Event studies

The analyses above indicate that prices changed at the time of the reform—yet it is possible that these changes were part of the ongoing evolution of the market. Moreover, the estimates above do not indicate whether there is additional adjustment beyond the first week towards the price adjustments predicted by our model. To address these issues, we conduct event studies for both the manufacturer and retailer responses using our baseline specifications from Tables 3 and 5. For manufacturers, we do not drop the $t-1$ tax reform coefficient due to our varied difference lengths.[21] Figure 4 plots the relevant coefficients and confidence intervals.[22]

Figure 4 demonstrates no discernible trend in prices pre-reform. This implies that once we control for the compositional shifts in Figure 1 with appropriate fixed effects, we no longer observe any significant trends in prices prior to the reform. The entire response happens in period t, the reform week. Given the varied difference lengths for manufacturers, this implies that manufacturers adjust their prices the first time they sell a particular retail-strain pair post-reform. This is compelling evidence that our estimates are unlikely to be driven by any ongoing market evolution and are instead a true response to the reform. The immediate nature of the response suggests that prices in this market follow a unit root process, further supporting our first-difference specification. Moreover, this suggests that our results are not driven by learning or other short-run adjustments. Appendix Figure A.3 provides longer-run evidence by repeating the manufacturer event study aggregated at the monthly level. We find no evidence of long-run learning either (Doraszelski et al., 2018; Huang et al., 2021).

[20] We could also estimate the response of retail margins to the tax reform directly on this sample to reach the same conclusion.

[21] E.g., for a 2-week difference that spans $t-1$ to $t+1$, both the t and $t+1$ coefficients are relevant.

[22] Appendix Figure A.2 replicates the event study plots 1 year later, further emphasizing the placebo findings in previous sections—our identification strategy is effective in this setting when tested in other periods with similar cyclicality and holiday patterns.

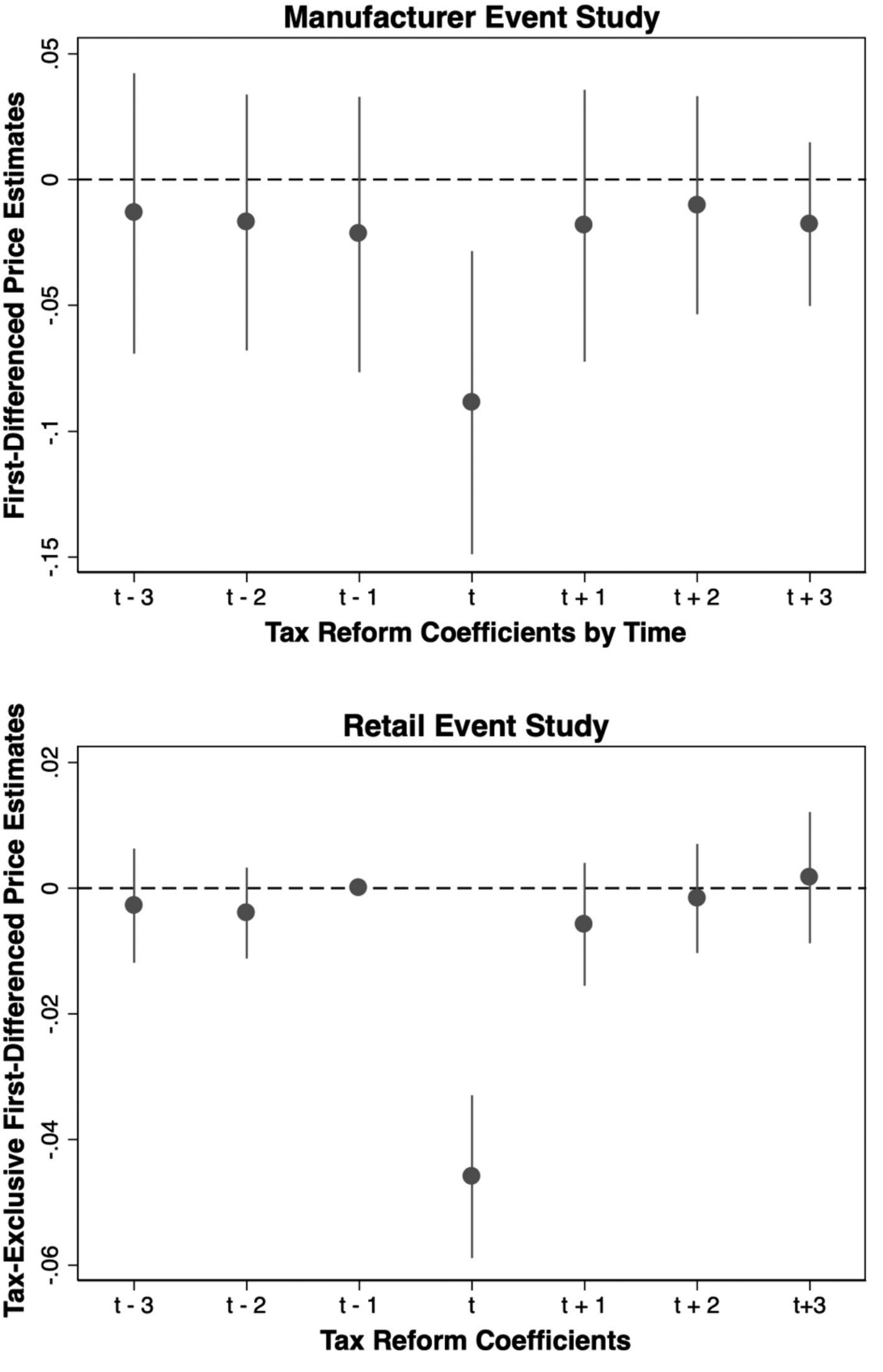

FIGURE 4 Manufacturer and retail price event studies.

Notes: This figure plots estimates of Table 3 column 2 (top panel) and Table 5 column 2 (bottom panel) with additional leads and lags of Δ*TaxReform*. The plotted coefficients are leads and lags of Δ*TaxReform*. We include in the regression (but do not plot) leads and lags are for periods $t-4$ and before and $t+4$ and after as is standard in event study designs. The dots indicate the point estimates and the lines indicate 95% confidence intervals. See the notes for Tables 3 and 5 for regression details.

DISCUSSION AND CONCLUSION

The public finance literature has documented several cases in which a change in tax rate or system changed the incidence of the tax on market participants. Firms don't capture Prius tax subsidies even though supply is inelastic (Sallee, 2011) and increases in value-added taxes are passed through to consumers at twice the rate of decreases (Benzarti et al., 2020). Prices don't respond to tax regime changes as predicted when evasion opportunities vary along the supply chain (Brockmeyer & Hernandez, 2016; Kopczuk et al., 2016; Slemrod, 2008), when prices are rigid (Lehmann et al., 2013; Muysken et al., 1999; Saez et al., 2012), or if tax salience differs between consumers and firms (Chetty et al., 2009; Finkelstein, 2009).

In each of these, unique market conditions including frictions or asymmetries likely contributed to the shift in incidence. We study a reform in a market with none of the issues previously documented and also find a substantial shift in incidence that had negative consequences for both retailers and consumers. This result is driven by the behavior of manufacturers, who on average reduced prices significantly less than a benchmark model of the industry predicted. Conditional on manufacturer prices, we find evidence that retailers applied a constant markup over marginal costs, consistent with standard models of firm behavior.[23]

We find evidence that these price changes were permanent in the sense that even 6 months later there were no industry-wide adjustments towards the expected equilibrium. To our knowledge, this reform remains the largest post-legalization reform of the cannabis industry in Washington state; while prices have continued to slowly decrease in the years after the reform, the industry structure and policy landscape has remained largely unchanged.

What explains this shift in incidence? The frequency of price changes—and the prevalence of at least some drop in manufacturer prices in response to the reform—suggest that managerial inattention is not relevant (Gabaix, 2019) and prices are not rigid (Lehmann et al., 2013). Our event studies suggest the response is immediate and Figure 1 does not find evidence of any longer-term response, which decreases the likelihood that learning can explain our findings. Unlike the setting of Benzarti et al. (2020), all members of the supply chain are likely aware of the tax reform and its implications on pricing.

We therefore view our results as consistent with models of behavioral phenomena, as opposed to models of information, transaction, or competitive frictions. In particular, anchoring and loss aversion may explain the outcomes we observe (Bernheim & Rangel, 2009; Kahneman et al., 1982, 1991). While the modal response by manufacturers in the week of the reform was to adjust their prices, the default option of "doing nothing" by maintaining constant tax-inclusive manufacturer prices (and thus realizing a significant increase in variable per-unit profits) may have anchored their negotiations with retail firms. The relatively common and somewhat small changes in manufacturer prices we do observe may be a result of competition—manufacturers may "do something" if they incorporate quantity or reputation effects into their analysis of post-tax outcomes (Rotemberg, 2011) and competitors may be compelled to act as a consequence. We also find evidence that negotiations were anchored by where manufacturers found themselves priced in the market prior to the reform; those with above average prices for a particular strain adjusted them downwards and those with below average prices adjusted them upwards in response to the tax reform. This decreased the overall price dispersion in the market.

Maintaining an anchor in the face of competition may have been easier for manufacturers as the immediacy and universality of the tax reform offered an opportunity for informal or implicit coordination among manufacturers, one that was perhaps easier to maintain than more typical settings where

[23] We expect our qualitative result that incidence shifted to be robust to a great deal of model uncertainty. In particular, if manufacturers employed average-cost pricing mechanisms (Altomonte et al., 2015; Hall & Hitch, 1939), we would expect the reform to cause similar or larger price drops than under marginal-cost pricing. While the reform eliminated incentives for inefficient vertical integration and, in the long run, production increased (Hansen et al., 2022), increased production efficiency should similarly drive down prices. In other words, reasonable alternative models of firm behavior would generate similar predictions and the same qualitative result.

price coordination often fails (Nakamura & Steinsson, 2013). While we can only speculate on private communications between firms, others have shown that firms can successfully coordinate through public communication alone (Aryal et al., 2022). There may be considerable complentarities between anchoring and loss averse behavior and the desire to coordinate: having an anchor to coalesce around may make it easier for multiple parties to coordinate, and the potential to coordinate may strengthen the anchor or the aversion to loss.[24]

Retailers, too, could be subject to anchoring and loss aversion. Why, then, did *they* change their prices in the aggregate? In the retailers' case, the "do nothing" option (maintaining constant tax-inclusive prices) represented a loss in variable per-unit profit. Once the default was overcome, they made decisions consistent with standard models.

We note that it is difficult to test this hypothesis explicitly and that other explanations are possible (though we have noted above key differences between our setting and the other examples of incidence-shifting documented by the literature). Zooming out, our results (and particularly the difference in responses between manufacturers and retailers) contribute to the growing literature in industrial organization exploring asymmetries in firm responses to changes in market conditions (Butters et al., 2022; DellaVigna & Gentzkow, 2019).

Our findings have important implications for tax policy, as they speak directly to the principle of tax invariance (TIV), which is a consequence of most classic models of taxation and states that who remits taxes does not influence who bears the burden of the tax.[25] TIV failed to hold in this setting, which notably lacks the frictions previously identified as responsible for other failures of TIV. In a world without TIV, designers of new taxes must consider any welfare tradeoffs involved in choosing where in a supply chain to assess a tax. Both efficiency and equity considerations arise. When considering efficiency, variation in elasticities or competitive structures across the market may affect optimal tax placement. In terms of equity, if a policy goal is to ensure all market participants bear portions of the tax, it may be necessary to impose taxes on these different groups directly.

Furthermore, changes to existing tax policy may generate first-order welfare consequences that were not predicted by classical models (Saez & Zucman, 2023). In our context, we find that restructuring a multistage or tiered tax system, even when intended to be universally welfare-enhancing, may instead create winners and losers. We find manufacturers benefited—despite being in an arguably more competitive market—while retailers and consumers were harmed. It is plausible that these results would generalize to other markets with multistage or tiered tax structure reforms where price coordination is difficult at baseline, which is common (Nakamura & Steinsson, 2013). Continuing to understand and model firm and consumer behavior in response to tax changes should be a central consideration in tax policy research and policy design moving forward.

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[24] In addition to losing some available profits, managers may feel badly about losing a rare opportunity to gain more profits.

[25] Modern models of competition, growth, trade, inflation, and the business cycle generally make assumptions about taxes which generally imply TIV (e.g., Galí, 2015; Judd, 2002; Melitz, 2003).

meetings, as well as industry participants and Cannabis Science and Policy Summit attendees. Many thanks to David Shi for excellent research assistance. This paper previously circulated as "Getting into the Weeds of Tax Invariance." Some of the results in this paper previously circulated as part of "The Taxation of Recreational Marijuana: Evidence from Washington State" and some of our thanks are for comments provided on that work. That paper no longer exists—all results were placed into other papers including this one.

ORCID

Caroline Weber https://orcid.org/0000-0001-7751-3523

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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AUTHOR BIOGRAPHIES

Benjamin Hansen is W.E. Miner Professor of Economics at University of Oregon, Eugene, OR 977403 (email: bchansen@uoregon.edu).

Kendall Houghton is an Economist at the Center for Economic Studies, U.S. Census Bureau (email: kendall.a.houghton@census.gov).

Keaton Miller is an Associate Professor of Economics at University of Oregon, Eugene, OR 977403 (email: keatonm@uoregon.edu).

Caroline Weber is an Associate Professor at Martin School of Public Policy & Administration, University of Kentucky, Lexington, KY 40506 (email: caroline.weber@uky.edu).